ML + Data Analytics
Exam Slide Priority Ranking

Based on the provided semester-final and quiz question pattern

How to read the stars: ★★★★★ = highest priority; ★★★☆☆ = lower priority among the supplied PDFs. Topics under each PDF are listed from most important to less important.

1. File Name: 8.pdf ★★★★★

Topic — Most important to less important

1. Linear Regression
2. Regression equation: slope, intercept, prediction
3. RMSE, MAE, R^2 / model evaluation
4. KNN calculation and classification
5. KNN: choice of K, noise, bias-variance, lazy learner
6. K-means clustering: distance, centroid, iterations
7. Classification vs clustering
8. Supervised vs unsupervised learning
9. Random Forest
10. Gradient Boosted Trees
11. Neural Networks

2. File Name: 7.pdf ★★★★★

Topic — Most important to less important

1. Training data vs testing data
2. Overfitting vs underfitting
3. Generalization and model performance
4. Structured vs unstructured vs semi-structured data
5. Data preprocessing
6. Data annotation
7. Image annotation
8. Role of training data in ML/AI

3. File Name: 4-Predictive-Aalysis.pdf ★★★★★

Topic — Most important to less important

1. Linear Regression

2. Model evaluation: R^2 and RMSE
3. Logistic Regression
4. Pandas DataFrame and basic data operations
5. Missing-value handling: mean, median, mode, fill/drop
6. Wrong-value correction and duplicate handling
7. Data visualization: scatter plot and histogram
8. OLAP / OLTP / business reporting

4. File Name: ML-2.pdf ★★★★★

Topic — Most important to less important

1. K-means clustering: algorithm and numerical steps
2. Supervised vs unsupervised learning
3. Clustering fundamentals
4. Distance calculation in clustering
5. Hierarchical clustering
6. Linear Regression
7. Gradient Descent
8. Association Rule Mining / Apriori
9. PCA / dimensionality reduction
10. DBSCAN
11. Gaussian Mixture Models
12. Reinforcement Learning

5. File Name: 6.pdf ★★★★☆

Topic — Most important to less important

1. Data mining tasks: prediction, association, clustering
2. Classification
3. Clustering
4. Supervised vs unsupervised learning
5. Major characteristics and objectives of data mining
6. Data mining process / CRISP-DM
7. KDD process
8. Data mining applications
9. Sequential relationships

6. File Name: 1-DA.pdf ★★★★☆

Topic — Most important to less important

1. Mean
2. Median
3. Quartiles / IQR
4. Variance
5. Standard Deviation
6. Range
7. Coefficient of Range
8. Frequency Distribution
9. Data Cleaning
10. Data Quality
11. Data Visualization / charts

7. File Name: 5.pdf ★★★★☆

Topic — Most important to less important

1. Data Mart
2. Dependent vs independent data mart
3. Data Warehouse definition
4. Data Warehouse characteristics: subject-oriented, integrated, time-variant, nonvolatile
5. Enterprise Data Warehouse (EDW)
6. Operational Data Store (ODS)
7. BI / Data Warehouse relationship

8. File Name: 2-DA.pdf ★★★★☆

Topic — Most important to less important

1. Correlation vs regression
2. Applications of correlation and regression
3. OLAP vs OLTP
4. Business Intelligence (BI)
5. Big Data definition
6. Normal distribution
7. Uniform distribution
8. Skewness
9. Evolution of BI / DSS / MIS

9. File Name: Chapter-8.pdf ★★★☆☆

Topic — Most important to less important

1. Predictive Analytics

2. Prescriptive Analytics
3. Descriptive Analytics
4. Diagnostic Analytics
5. Predictive vs prescriptive comparison
6. Big Data definition
7. Characteristics of Big Data: Volume, Velocity, Variety, Veracity, Value
8. Big Data Analytics flow
9. Big Data examples / domains
10. Seven Giants

10. File Name: ML-3.pdf ★★★☆☆

Topic — Most important to less important

1. Overfitting vs underfitting
2. Bias vs variance
3. Bias-variance tradeoff
4. Regularization
5. Gradient Descent
6. Batch Gradient Descent
7. Stochastic Gradient Descent (SGD)
8. Mini-batch Gradient Descent
9. Learning rate

Final focus order: Regression → KNN → K-means → model evaluation → preprocessing/data cleaning → training/testing → data mining → data mart/BI → analytics types → remaining ML topics.